

2014 January Q10 C34

10. With respect to a fixed origin O , the lines l_1 and l_2 are given by the equations

$$l_1 : \mathbf{r} = (\mathbf{i} + 5\mathbf{j} + 5\mathbf{k}) + \lambda(2\mathbf{i} + \mathbf{j} - \mathbf{k})$$

$$l_2 : \mathbf{r} = (2\mathbf{j} + 12\mathbf{k}) + \mu(3\mathbf{i} - \mathbf{j} + 5\mathbf{k})$$

where λ and μ are scalar parameters.

(a) Show that l_1 and l_2 meet and find the position vector of their point of intersection. (6)

(b) Show that l_1 and l_2 are perpendicular to each other. (2)

The point A , with position vector $5\mathbf{i} + 7\mathbf{j} + 3\mathbf{k}$, lies on l_1

The point B is the image of A after reflection in the line l_2

(c) Find the position vector of B . (3)

2014 June Q14 C34

14. Relative to a fixed origin O , the line l has vector equation

$$\mathbf{r} = \begin{pmatrix} -1 \\ -4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

where λ is a scalar parameter.

Points A and B lie on the line l , where A has coordinates $(1, a, 5)$ and B has coordinates $(b, -1, 3)$.

(a) Find the value of the constant a and the value of the constant b .

(3)

(b) Find the vector $\overrightarrow{AB}$.

(2)

The point C has coordinates $(4, -3, 2)$

(c) Show that the size of the angle CAB is 30°

(3)

(d) Find the exact area of the triangle CAB , giving your answer in the form $k\sqrt{3}$, where k is a constant to be determined.

(2)

The point D lies on the line l so that the area of the triangle CAD is twice the area of the triangle CAB .

(e) Find the coordinates of the two possible positions of D .

(4)

2015 January Q11 C34

11. With respect to a fixed origin O the lines l_1 and l_2 are given by the equations

$$l_1: \mathbf{r} = \begin{pmatrix} 14 \\ -6 \\ -13 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} \quad l_2: \mathbf{r} = \begin{pmatrix} p \\ -7 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} q \\ 2 \\ 1 \end{pmatrix}$$

where λ and μ are scalar parameters and p and q are constants.

Given that l_1 and l_2 are perpendicular,

(a) show that $q = 3$

(2)

Given further that l_1 and l_2 intersect at point X ,

find

(b) the value of p ,

(5)

(c) the coordinates of X .

(2)

The point A lies on l_1 and has position vector $\begin{pmatrix} 6 \\ -2 \\ 3 \end{pmatrix}$

Given that point B also lies on l_1 and that $AB = 2AX$

(d) find the two possible position vectors of B .

(3)

4.

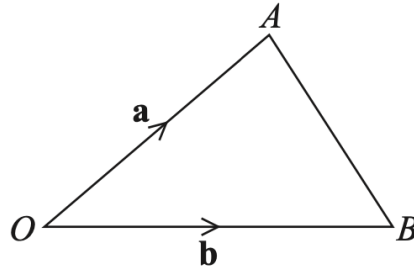


Figure 2

Figure 2 shows the points A and B with position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, relative to a fixed origin O .

Given that $|\mathbf{a}| = 5$, $|\mathbf{b}| = 6$ and $\mathbf{a} \cdot \mathbf{b} = 20$

- (a) find the cosine of angle AOB , (2)
- (b) find the exact length of AB . (2)
- (c) Show that the area of triangle OAB is $5\sqrt{5}$ (3)

2015 June Q12 C34

12. (i) Relative to a fixed origin O , the line l_1 is given by the equation

$$l_1: \mathbf{r} = \begin{pmatrix} -5 \\ 1 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \text{ where } \lambda \text{ is a scalar parameter.}$$

The point P lies on l_1 . Given that $\vec{OP}$ is perpendicular to l_1 , calculate the coordinates of P .

(5)

(ii) Relative to a fixed origin O , the line l_2 is given by the equation

$$l_2: \mathbf{r} = \begin{pmatrix} 4 \\ -3 \\ 12 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ -3 \\ 4 \end{pmatrix} \text{ where } \mu \text{ is a scalar parameter.}$$

The point A **does not** lie on l_2 . Given that the vector $\vec{OA}$ is parallel to the line l_2 and $|\vec{OA}| = \sqrt{2}$ units, calculate the possible position vectors of the point A .

(5)

2016 January Q12 C34

12. With respect to a fixed origin O , the lines l_1 and l_2 are given by the equations

$$l_1 : \mathbf{r} = \begin{pmatrix} 12 \\ -4 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -4 \\ 2 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 6 \\ 3 \end{pmatrix}$$

where λ and μ are scalar parameters.

(a) Show that l_1 and l_2 meet, and find the position vector of their point of intersection A .
(6)

(b) Find, to the nearest 0.1° , the acute angle between l_1 and l_2
(3)

The point B has position vector $\begin{pmatrix} 7 \\ 0 \\ 3 \end{pmatrix}$.

(c) Show that B lies on l_1
(1)

(d) Find the shortest distance from B to the line l_2 , giving your answer to 3 significant figures.
(4)

2016 June Q11 C34

11. With respect to a fixed origin O , the lines l_1 and l_2 are given by the equations

$$l_1 : \mathbf{r} = \begin{pmatrix} 7 \\ 4 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -6 \\ -7 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 4 \\ b \end{pmatrix}$$

where λ and μ are scalar parameters and b is a constant.

Given that l_1 and l_2 meet at the point X ,

(a) show that $b = -3$ and find the coordinates of X .

(5)

The point A lies on l_1 and has coordinates $(6, 3, 5)$

The point B lies on l_2 and has coordinates $(14, 9, -9)$

(b) Show that angle $AXB = \arccos\left(-\frac{1}{10}\right)$

(4)

(c) Using the result obtained in part (b), find the exact area of triangle AXB .

Write your answer in the form $p\sqrt{q}$ where p and q are integers to be determined.

(3)

2017 January Q14 C34

14. $ABCD$ is a parallelogram with AB parallel to DC and AD parallel to BC .

The position vectors of A , B , C , and D relative to a fixed origin O are $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ respectively.

Given that

$$\mathbf{a} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}, \quad \mathbf{b} = 3\mathbf{i} - \mathbf{j} + 6\mathbf{k}, \quad \mathbf{c} = -\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$$

(a) find the position vector $\mathbf{d}$,

(3)

(b) find the angle between the sides AB and BC of the parallelogram,

(4)

(c) find the area of the parallelogram $ABCD$.

(2)

The point E lies on the line through the points C and D , so that D is the midpoint of CE .

(d) Use your answer to part (c) to find the area of the trapezium $ABCE$.

(2)

2017 June Q6 C34

6. The line l_1 has vector equation $\mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 3 \\ -1 \end{pmatrix}$, where λ is a scalar parameter.

The line l_2 has vector equation $\mathbf{r} = \begin{pmatrix} 10 \\ 5 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$, where μ is a scalar parameter.

Justify, giving reasons in each case, whether the lines l_1 and l_2 are parallel, intersecting or skew.

(6)

9.

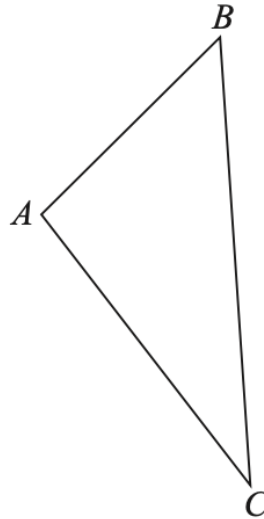


Figure 2

Figure 2 shows a sketch of a triangle ABC .

Given $\vec{AB} = 2\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ and $\vec{AC} = 5\mathbf{i} - 6\mathbf{j} + \mathbf{k}$,

(a) find the size of angle CAB , giving your answer in degrees to 2 decimal places, (3)

(b) find the area of triangle ABC , giving your answer to 2 decimal places. (2)

Using your answer to part (b), or otherwise,

(c) find the shortest distance from A to BC , giving your answer to 2 decimal places. (3)

2017 October Q12 C34

12. Relative to a fixed origin O , the lines l_1 and l_2 are given by the equations

$$l_1: \mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \qquad l_2: \mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} 8 \\ 4 \\ 1 \end{pmatrix}$$

where λ and μ are scalar parameters.

The lines l_1 and l_2 intersect at the point A .

(a) Write down the coordinates of A .

(1)

Given that the acute angle between l_1 and l_2 is θ ,

(b) show that $\sin \theta = k\sqrt{2}$, where k is a rational number to be found.

(5)

The point B lies on l_1 where $\lambda = 4$

The point C lies on l_2 such that $AC = 2AB$.

(c) Find the exact area of triangle ABC .

(3)

(d) Find the coordinates of the two possible positions of C .

(5)

2018 January Q7 C34

7. With respect to a fixed origin O , the lines l_1 and l_2 are given by the equations

$$l_1: \mathbf{r} = (13\mathbf{i} + 15\mathbf{j} - 8\mathbf{k}) + \lambda(3\mathbf{i} + 3\mathbf{j} - 4\mathbf{k})$$

$$l_2: \mathbf{r} = (7\mathbf{i} - 6\mathbf{j} + 14\mathbf{k}) + \mu(2\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})$$

where λ and μ are scalar parameters.

- (a) Show that l_1 and l_2 meet and find the position vector of their point of intersection, B .
(6)

- (b) Find the acute angle between the lines l_1 and l_2
(3)

The point A has position vector $-5\mathbf{i} - 3\mathbf{j} + 16\mathbf{k}$

- (c) Show that A lies on l_1
(1)

The point C lies on the line l_1 where $\overrightarrow{AB} = \overrightarrow{BC}$

- (d) Find the position vector of C .
(3)

2018 June Q8 C34

8. With respect to a fixed origin O , the lines l_1 and l_2 are given by the equations

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 6 \\ 4 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

where λ and μ are scalar parameters.

- (a) Show that l_1 and l_2 do not meet.

(4)

The point P is on l_1 where $\lambda = 0$, and the point Q is on l_2 where $\mu = -1$

- (b) Find the acute angle between the line segment PQ and l_1 , giving your answer in degrees to 2 decimal places.

(5)

- (c) Find the shortest distance from the point Q to the line l_1 , giving your answer to 3 significant figures.

(2)

2018 October Q11 C34

11. Relative to a fixed origin O , the line l_1 is given by the equation

$$l_1: \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix}$$

where λ is a scalar parameter.

The line l_2 passes through the origin and is parallel to l_1

(a) Find a vector equation for l_2

(2)

The point A and the point B both lie on l_1 with parameters $\lambda = 0$ and $\lambda = 3$ respectively.

Write down

(b) (i) the coordinates of A ,

(ii) the coordinates of B .

(2)

(c) Find the size of the acute angle between OA and l_1

Give your answer in degrees to one decimal place.

(3)

The point D lies on l_2 such that $OABD$ is a parallelogram.

(d) Find the area of $OABD$, giving your answer to the nearest whole number.

(3)

2019 January Q6 C34

6. Relative to a fixed origin O , the points A , B and C have coordinates $(2, 1, 9)$, $(5, 2, 7)$ and $(4, -3, 3)$ respectively.

The line l passes through the points A and B .

- (a) Find a vector equation for the line l .

(2)

- (b) Find, in degrees, the acute angle between the line l and the line AC .

(3)

The point D lies on the line l such that angle ACD is 90°

- (c) Find the coordinates of D .

(4)

- (d) Find the exact area of triangle ADC , giving your answer as a fully simplified surd.

(2)

2019 June Q12 C34

12. Relative to a fixed origin O ,

the point A has position vector $(2\mathbf{i} - 3\mathbf{j} - 2\mathbf{k})$

the point B has position vector $(3\mathbf{i} + 2\mathbf{j} + 5\mathbf{k})$

the point C has position vector $(2\mathbf{i} + 4\mathbf{j} - 3\mathbf{k})$

The line l passes through the points A and B .

(a) Find the vector $\overrightarrow{AB}$. (2)

(b) Find a vector equation for the line l . (2)

(c) Show that the size of the angle CAB is 62.8° , to one decimal place. (4)

(d) Hence find the area of triangle CAB , giving your answer to 3 significant figures. (2)

The point D lies on the line l . Given that the area of triangle CAD is twice the area of triangle CAB ,

(e) find the two possible position vectors of point D . (3)

2020 October Q8 P4

8. Relative to a fixed origin O , the lines l_1 and l_2 are given by the equations

$$l_1: \mathbf{r} = \begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix} \quad \text{where } \lambda \text{ is a scalar parameter}$$

$$l_2: \mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ -9 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ -3 \end{pmatrix} \quad \text{where } \mu \text{ is a scalar parameter}$$

Given that l_1 and l_2 meet at the point X ,

(a) find the position vector of X .

(5)

The point $P(10, -7, 0)$ lies on l_1

The point Q lies on l_2

Given that $\vec{PQ}$ is perpendicular to l_2

(b) calculate the coordinates of Q .

(5)

2.

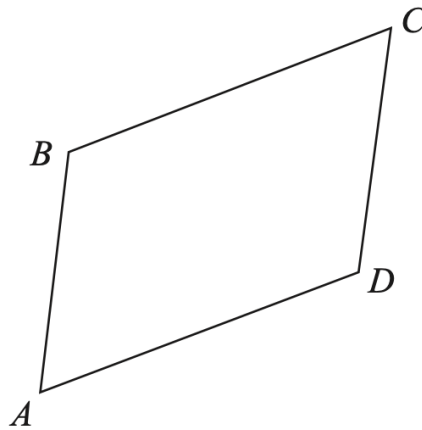


Figure 1

Figure 1 shows a sketch of parallelogram $ABCD$.

Given that $\vec{AB} = 6\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\vec{BC} = 2\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}$

(a) find the size of angle ABC , giving your answer in degrees, to 2 decimal places.

(3)

(b) Find the area of parallelogram $ABCD$, giving your answer to one decimal place.

(2)

2021 January Q8 P4

8. With respect to a fixed origin O , the lines l_1 and l_2 are given by the equations

$$l_1: \mathbf{r} = \begin{pmatrix} -1 \\ 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} \qquad l_2: \mathbf{r} = \begin{pmatrix} 2 \\ -2 \\ -5 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -3 \\ b \end{pmatrix}$$

where λ and μ are scalar parameters and b is a constant.

Prove that for all values of $b \neq 7$, the lines l_1 and l_2 are skew.

(6)

2021 June Q7 P4

7. Relative to a fixed origin O , the line l has equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ -10 \\ -9 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 4 \\ 2 \end{pmatrix} \quad \text{where } \lambda \text{ is a scalar parameter}$$

Given that $\vec{OA}$ is a unit vector parallel to l ,

(a) find $\vec{OA}$ (2)

The point X lies on l .

Given that X is the point on l that is closest to the origin,

(b) find the coordinates of X . (5)

The points O , X and A form the triangle OXA .

(c) Find the exact area of triangle OXA . (3)

2021 October Q7 P4

7. With respect to a fixed origin O ,

- the line l has equation $\mathbf{r} = \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ -3 \\ 5 \end{pmatrix}$ where λ is a scalar constant
- the point A has position vector $9\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$

Given that X is the point on l nearest to A ,

(a) find

- the coordinates of X
- the shortest distance from A to l .

Give your answer in the form $\sqrt{d}$, where d is an integer.

(7)

The point B is the image of A after reflection in l .

(b) Find the position vector of B .

(2)

2022 January Q8 P4

8. With respect to a fixed origin O the points A and B have position vectors

$$\begin{pmatrix} 6 \\ 6 \\ 2 \end{pmatrix} \text{ and } \begin{pmatrix} 6 \\ 0 \\ 7 \end{pmatrix}$$

respectively.

The line l_1 passes through the points A and B .

- (a) Write down an equation for l_1

Give your answer in the form $\mathbf{r} = \mathbf{p} + \lambda \mathbf{q}$, where λ is a scalar parameter.

(2)

The line l_2 has equation

$$\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 5 \\ 9 \end{pmatrix}$$

where μ is a scalar parameter.

- (b) Show that l_1 and l_2 do **not** meet.

(4)

The point C is on l_2 where $\mu = -1$

- (c) Find the acute angle between AC and l_2

Give your answer in degrees to one decimal place.

(5)

2022 June Q6 P4

6. Relative to a fixed origin O ,

- the point A has position vector $\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}$
- the point B has position vector $5\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$
- the point C has position vector $3\mathbf{i} + p\mathbf{j} - \mathbf{k}$

where p is a constant.

The line l passes through A and B .

(a) Find a vector equation for the line l

(3)

Given that $\overrightarrow{AC}$ is perpendicular to l

(b) find the value of p

(3)

(c) Hence find the area of triangle ABC , giving your answer as a surd in simplest form.

(3)

3.

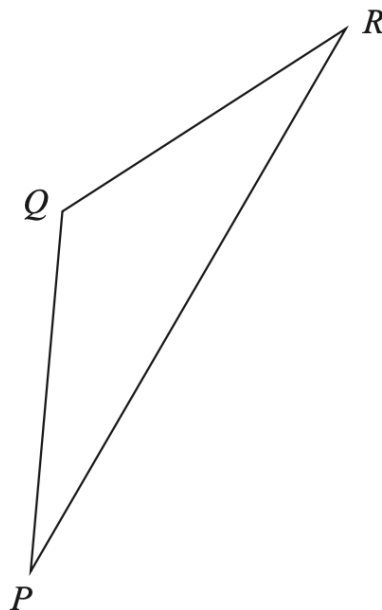


Figure 1

Figure 1 shows a sketch of triangle PQR .

Given that

- $\vec{PQ} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$
- $\vec{PR} = 8\mathbf{i} - 5\mathbf{j} + 3\mathbf{k}$

(a) Find $\vec{RQ}$

(2)

(b) Find the size of angle PQR , in degrees, to three significant figures.

(3)

2022 October Q9 P4

9. With respect to a fixed origin O , the equations of lines l_1 and l_2 are given by

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 8 \\ 10 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}$$

$$l_2 : \mathbf{r} = \begin{pmatrix} -4 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 4 \\ 8 \end{pmatrix}$$

where λ and μ are scalar parameters.

Prove that lines l_1 and l_2 are skew.

(5)

2023 January Q6 P4

6. Relative to a fixed origin O .

- the point A has position vector $2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$
- the point B has position vector $8\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}$

The line l passes through A and B .

(a) (i) Find $\overrightarrow{AB}$

(ii) Find a vector equation for the line l

(3)

The point C has position vector $3\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}$

The point P lies on l

Given that $\overrightarrow{CP}$ is perpendicular to l

(b) find the position vector of the point P

(5)

2023 June Q4 P4

4. Relative to a fixed origin O ,

- the point A has position vector $4\mathbf{i} + 8\mathbf{j} + \mathbf{k}$
- the point B has position vector $5\mathbf{i} + 6\mathbf{j} + 3\mathbf{k}$
- the point P has position vector $2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

The straight line l passes through A and B .

(a) Find a vector equation for l .

(2)

The point C lies on l so that PC is perpendicular to l .

(b) Find the coordinates of C .

(4)

The point P' is the reflection of P in the line l .

(c) Find the coordinates of P'

(2)

(d) Hence find $|\overrightarrow{PP'}|$, giving your answer as a simplified surd.

(2)

2023 October Q6 P4

6. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -7 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ where λ is a scalar parameter.

The line l_2 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -1 \\ 8 \end{pmatrix}$ where μ is a scalar parameter.

Given that l_1 and l_2 meet at the point P

(a) state the coordinates of P (1)

Given that the angle between lines l_1 and l_2 is θ

(b) find the value of $\cos \theta$, giving the answer as a fully simplified fraction. (3)

The point Q lies on l_1 where $\lambda = 6$

Given that point R lies on l_2 such that triangle QPR is an isosceles triangle with $PQ = PR$

(c) find the exact area of triangle QPR (3)

(d) find the coordinates of the possible positions of point R (3)

2024 January Q6 P4

6. Relative to a fixed origin O , the lines l_1 and l_2 are given by the equations

$$l_1 : \mathbf{r} = (3\mathbf{i} + p\mathbf{j} + 7\mathbf{k}) + \lambda(2\mathbf{i} - 5\mathbf{j} + 4\mathbf{k})$$

$$l_2 : \mathbf{r} = (8\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}) + \mu(4\mathbf{i} + \mathbf{j} + 2\mathbf{k})$$

where λ and μ are scalar parameters and p is a constant.

Given that l_1 and l_2 intersect,

(a) find the value of p , (4)

(b) find the position vector of the point of intersection. (2)

(c) Find the acute angle between l_1 and l_2
Give your answer in degrees to one decimal place. (3)

The point A lies on l_1 with parameter $\lambda = 2$

The point B lies on l_2 with $\overrightarrow{AB}$ perpendicular to l_2

(d) Find the coordinates of B (5)